Tense Timing Experiments: Analysis & Report

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Models of language production treat morphological features with a good deal of uniformity. In the lemma model, morphosyntactic information is carried by diacritics on lemma nodes, and the mechanism that assigns a diacritic is much the same whether it encodes number, tense or case. Cue-based models are architecturally different but inherit the same lexicalist premise: features are properties of chunks retrieved from memory, and nothing in the architecture distinguishes one kind of feature from another in its timing. Both make morphological encoding dependent on lexical selection, and therefore subsequent to it. A feature whose information source is available early is nonetheless held behind the same bottleneck as one that is not.

Tense is a good place to test that premise, because its information source is unusually early. Agreement reaches the verb through a linguistic mediator: the discourse fixes the number of the subject, and the subject fixes the number of the verb, so the verb's number cannot be specified until another element has been planned. Tense has no such mediator. The speaker's communicative intention fixes whether the event is past or future, and that is known from the very start of planning. If the production system can specify a morphosyntactic feature without first retrieving the word that carries it, tense is where it should be visible.

That makes it worth separating two things usually spoken about in one breath. One is specification of the tense feature itself, marking the relevant node as past or future as a function of the discourse. The other is retrieval of the form that realises it, *spun* against *will spin*, which cannot be settled without knowing which verb is being produced, since the shape of the past depends on the verb's lexical identity. The second cannot precede lexical retrieval. The first, on the account pursued here, can.

The three experiments isolate the levels in turn. Experiment 1 holds the task constant and repeats the tense, so an effect early in the utterance would show the feature being specified before the verb has been reached. Experiment 2 holds tense constant, every event being past, and repeats the way the past is formed instead; here an effect is expected late, around the verb. The contrast with agreement is instructive: in agreement two lexical items bear on one closed-class element, whereas here a single piece of information, past, meets a verb whose lexical identity refracts it into a regular or an irregular surface form. Experiment 3 repeats the same decision about time with no utterance at all, and so tests whether a repetition effect needs linguistic material to act on.

The two accounts differ in where they place these operations. Figure 1 lays them over the utterance. Under the first, the tense feature is settled before articulation begins and leaves its trace at the left, while the form is retrieved only as the verb is approached and leaves its trace at the right. Under the second, the diacritic waits until the word carrying it is about to be spoken, so both fall together in the pause before the verb and neither leaves a trace earlier. The accounts agree from the verb onwards, so the determiner and the subject noun are where they can be told apart.

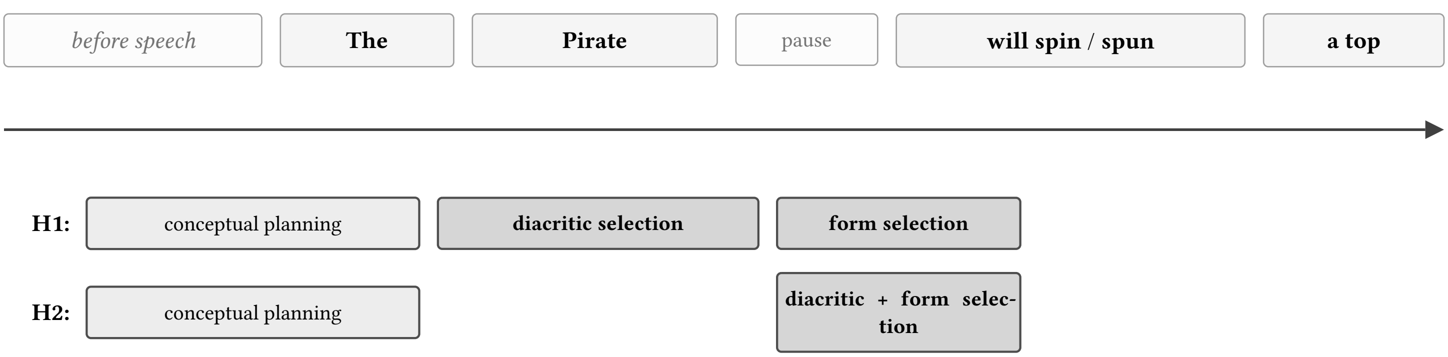


Figure 1: Two accounts of when tense is planned, laid over the utterance they predict about. Time runs left to right. Under H1 the tense feature is fixed early, while the sentence is still being begun, and the form that spells it out is retrieved only as the verb is approached. Under H2 both are resolved together, late, in the pause before the verb and just into it. The accounts make the same predictions from the verb onwards and differ only over the determiner and the subject noun, which is where they can be told apart.

All timing measures are in milliseconds. Error bars are 95% intervals built on Morey (2008) within-subject standard errors. Instead of reporting statistical significance, we report the posterior mean, the 95% credible interval, and the proportion of the posterior falling on one side of zero. An effect is called *credible* when the interval excludes zero, *leaning* when at least 85% of the posterior is on one side but the interval still spans zero, and *flat* otherwise.

1 Data and preprocessing

Experiments were carried out in PcIbex, and spoken responses were uploaded per block as `.webm`, converted to 16 kHz mono wav, transcribed with AssemblyAI, and force-aligned against those transcripts with the Montreal Forced Aligner (`english_us_arpa` acoustic model and dictionary, out-of-vocabulary items passed through G2P). Word and phone¡™ boundaries come from the resulting TextGrids. The commands are listed in the appendix. Recordings that never reached the server, and sessions with no usable audio, are excluded and will not reported in any analysis below.

1.1 Participant filtering

Forced alignment yielded 48 morphophonology participants and 63 morphosyntax participants. Participants who passed fewer than 50% of trials on the utterance criteria were excluded: 7 of 48 in morphophonology (41 remaining), and 17 of 63 in morphosyntax (46 remaining). After all participant-level exclusions, 41 morphophonology participants and 46 morphosyntax participants remain.

Within the retained participants, individual trials were excluded when the response did not name the correct subject entity, did not use the target verb in the tense the trial called for, did not match the target sentence in length, was empty, or marked tense with an adverbial instead of on the verb. This retained 3799 of 4344 trials (87%) in morphophonology and 3921 of 4947 trials (79%) in morphosyntax.

2 Descriptive results

The condition means for all three experiments are in Figure 2. Experiments 1 and 2 share their five measures, so their rows are directly comparable; Experiment 3 has only the keypress latency, and its single panel is drawn to the width of one panel in the rows above it. The tables behind these figures are in the appendix.

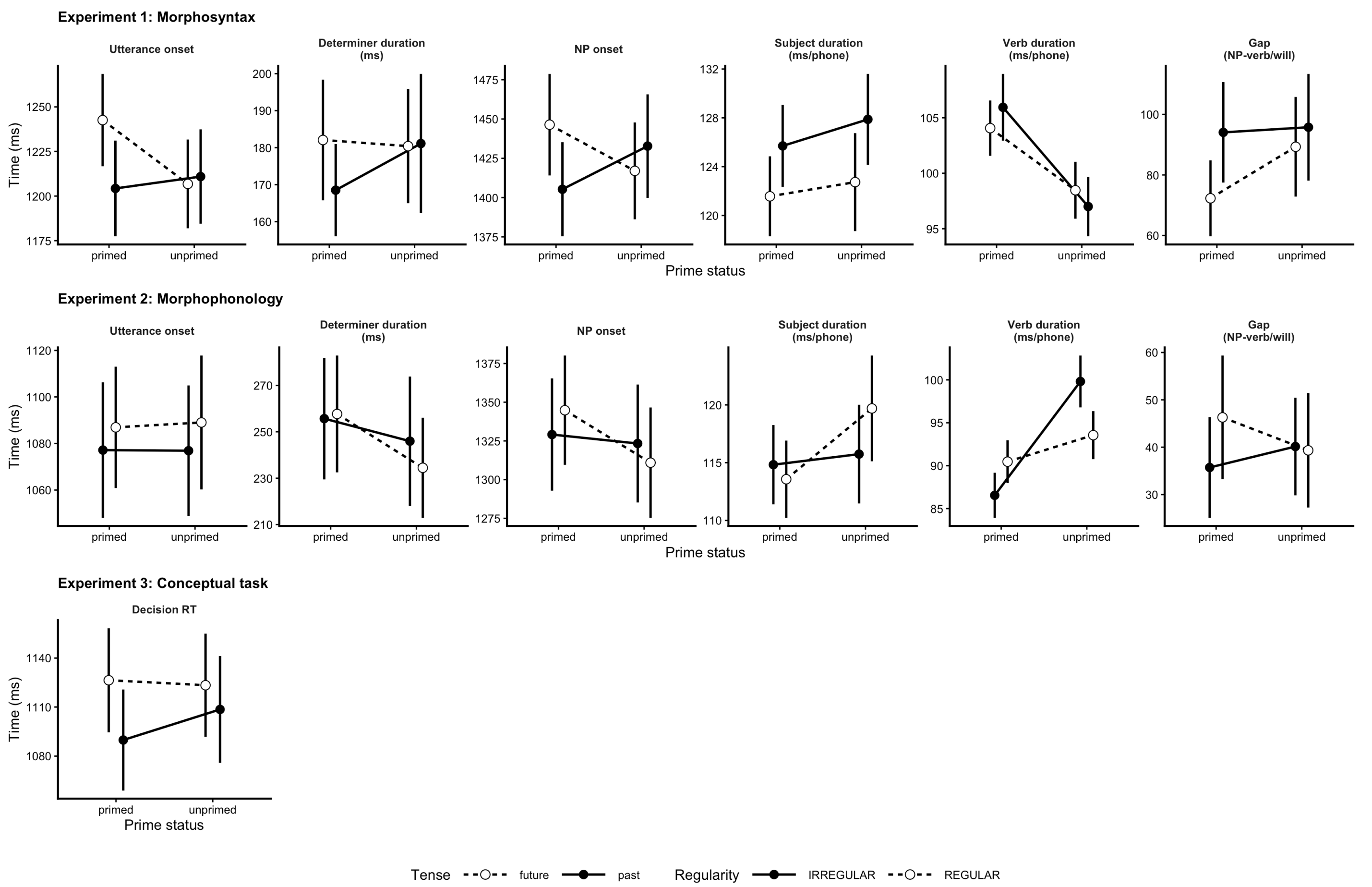


Figure 2: Condition means for all three experiments. Top row, Experiment 1: timing by tense and tense repetition. Middle row, Experiment 2: timing by regularity and regularity repetition. Bottom row, Experiment 3: decision latency by tense and tense repetition, drawn at the width of a single panel above. Onsets and the gap are in milliseconds, the two durations in milliseconds per phoneme, and the decision latency in milliseconds. Filled markers are the primed level, open markers its baseline. Error bars are 95% intervals on Morey (2008) within-subject standard errors.

3 Experiment 1: Morphosyntax

Producing a tensed sentence requires two decisions that can come apart. One is which tense the sentence carries, a choice about its structure. The other is which form realises that tense on the verb. If the two are separate operations they should be able to leave their traces at different points in the utterance. This experiment isolates the first: participants describe the same scenes in the past or the future, and what is modulated whether the tense of the trial is being repeated from trial to trial.

3.1 Prediction

The effects related to the repetition should surface early in the sentence. Choosing between past and future is a choice about the structure of the sentence, possibly made before articulation begins, so its effects should appear at the start of the utterance rather than around the verb. The interaction between tense and tense repetition at the onset of the subject noun is the measure this predicts, and the later measures should show little.

3.2 Participants

46 participants are analysed here. All were recruited through the University of Maryland SONA subject pool and received course credit. Age, gender and language background were collected at the start of the session.

The study was approved by the University of Maryland Institutional Review Board. Participants read an information sheet describing the task, the recording of their voice, and their right to stop at any point without penalty, and gave informed consent before beginning. Responses are identified only by the pool identifier needed to award credit and by a per-session token; no names or contact details were collected, and the recordings and results analysed here carry no information that identifies a participant directly.

3.3 Materials

Three characters appear throughout: the Pirate, the Chef and the Wizard. Eighteen verbs are divided into three sets of six, one set per block:

Block	Verbs
1	drink, read, eat, paint, wash, push
2	build, sweep, ride, climb, stir, peel
3	blow, dig, shake, carry, play, smell

Two further verbs, *spin* and *drag*, are held back for practice and never used in the main experiment.

Each block, first, familiarizes the event with the participant without the exact sentence to utter and without either the past or future tense marking:

(1) The Pirate's spinning a top is in the past.

Depending on the experimental condition, participants either produced past or the future marked information.

(2) a. The Pirate spun a top. (past)
b. The Pirate will spin a top. (future)

3.4 Procedure

After consent, demographics and a microphone check, participants read the instructions, learned two practice verbs and produced two practice sentences.

The main task is built from nine blocks, arranged in three meta-blocks of three. A block is the unit that pairs learning with production. It opens with a familiarisation phase: the six verbs of that block are presented one at a time with audio, and then each is placed in time, so the participant learns both the event and whether it is being described as past or future. Only after that phase does the participant produce anything. The twelve production trials that follow show a picture and ask for a spoken description, with no cue or reminder of the tense on the screen; what the participant produces comes from what they learned at the start of the block. Breaks fall between blocks, and the second and third meta-blocks open with a screen announcing that the events now belong to a new situation, because the same verbs recur with different characters and different times.

Nine blocks of twelve trials give 108 recorded sentences per participant. The eighteen verbs are covered once per meta-block, so each verb is produced six times over the session.

Figure 3 lays out the session, one block, and one trial.

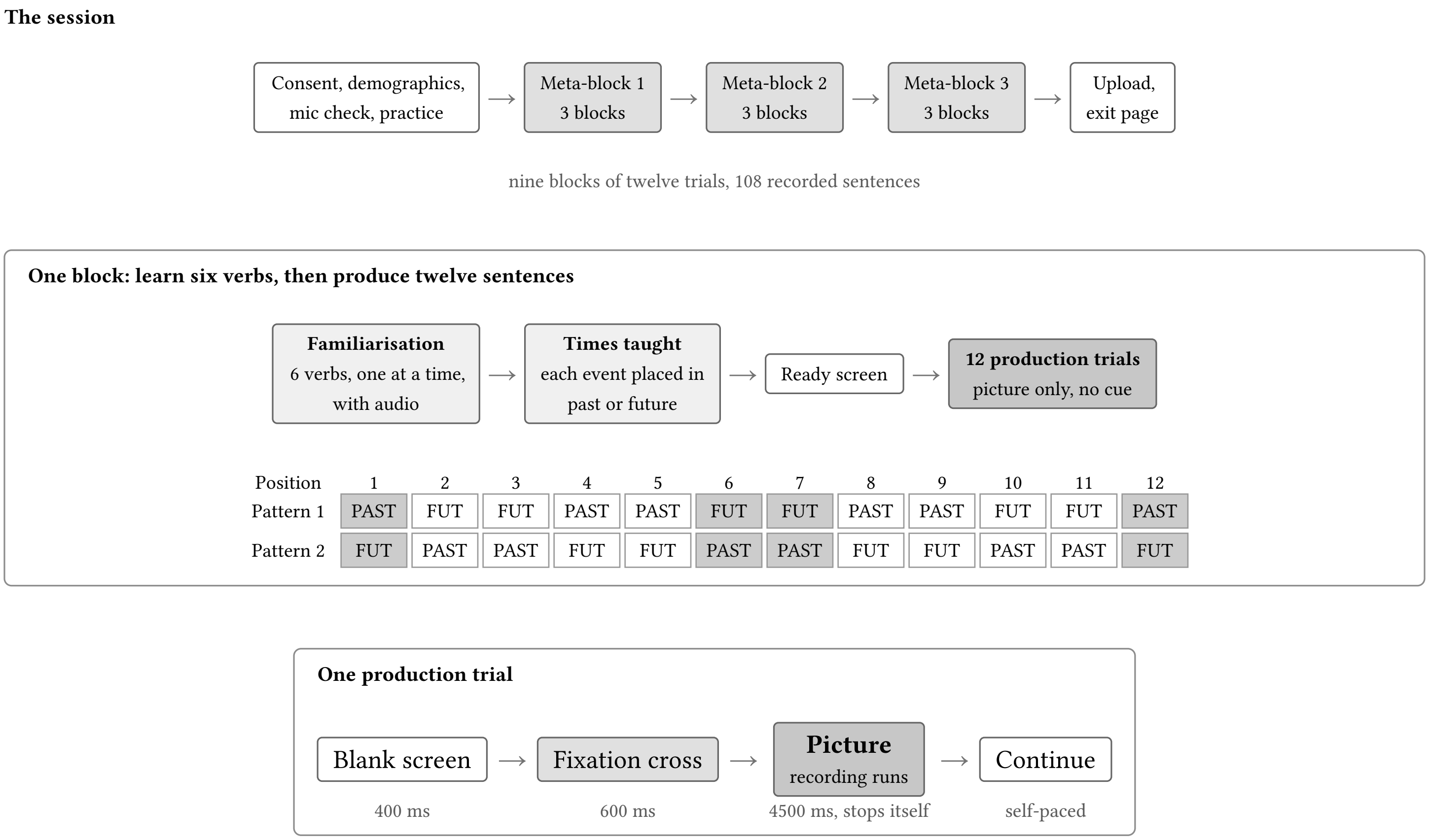


Figure 3: Procedure for Experiment 1. Learning and production are separate phases of a block, not steps within a trial: the six verbs of a block are taught first, with audio and with each event placed in past or future, and the twelve production trials that follow show only a picture, with nothing on the screen naming the tense. Those twelve trials are two passes over the block's six verbs, one under each tense order, with the order of the passes randomised; shaded positions are boundary trials, dropped because they have no comparable predecessor, and every other trial is primed when it repeats the tense of the trial before it and unprimed when it does not. Experiment 2 has the same shape with regularity in place of tense and no time-teaching step, and Experiment 3 replaces the spoken response with a keypress.

Which pattern runs first is randomised per block, block order is shuffled within each meta-block, the assignment of characters to verbs rotates across meta-blocks, and the ordering avoids placing the same character on consecutive trials.

3.5 Analysis

Five measures are taken from the forced alignment of each recording: utterance onset, the onset of the subject noun, the per-phoneme duration of the subject noun, the per-phoneme duration of the verb, and the gap running from the offset of the subject noun to the tense-bearing word, which is *will* in the future and the past-marked verb in the past. Experiment 2 uses the same five measures; Experiment 3 has no audio and measures keypress latency instead.

Each measure is modelled separately. The terms of interest are tense, tense repetition, and their interaction. Repetition of the morphophonological form is carried alongside as a control, because a trial that repeats the tense often repeats the form as well. Frequency, the subject noun and articulation rate are also controlled. The appendix gives the contrast coding, the priors, the full random-effect structure and the convergence checks.

Of 3921 trials, 1639 were boundary trials that was not included in the data, leaving **2661 trials** from 46 participants.

The model estimates each measure separately. The measures are the time from the start of the trial to the first word, the time to the subject noun, the duration of the subject noun per phone, the duration of the verb per phone, and the interval between the subject noun and the verb.

How to read the signs. A positive coefficient means longer or later onsets and durations; a negative one means shorter or earlier. Predictors are sum coded at ±0.5, so each estimate is the difference between its two levels: Tense +0.5 = future, −0.5 = past; Tense priming +0.5 = tense-primed, −0.5 = tense-unprimed. An interaction is the difference between two such differences.

Speakers reach the verb without pausing on most trials, so the interval is modelled in two parts. **Gap length** is how long the pause lasts on the trials where there is one. **No pause lik.** is how likely the speaker is to run straight into the verb; a positive coefficient there means fewer pauses. The first part is on the log scale in milliseconds, the second on the log-odds scale.

3.6 Results

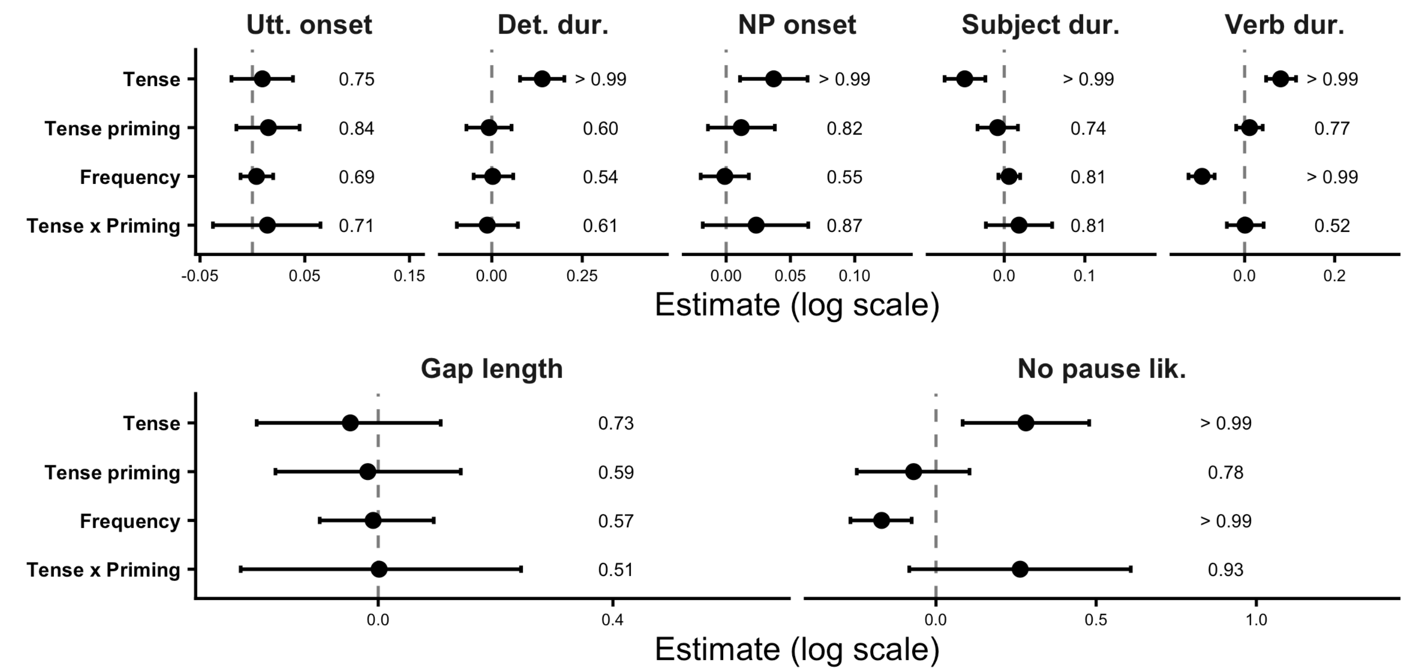


Figure 4: Experiment 1: population-level coefficients of the reported model. Points are posterior medians, bars 95% credible intervals; the dashed line marks zero. The figure at the right of each panel is the proportion of the posterior on one side of zero.

Tense itself changed the shape of the utterance from the determiner onwards (Figure 2, top row). Speakers took longer over the determiner before a future sentence than before a past one, 181 ms (SE 5.4) against 175 ms (SE 5.3), and the model confirms the difference (β = 0.139, 95% CrI [0.078, 0.201], P(dir) = > 0.99). This is the earliest point at which tense could show up at all, and *the* is the same word in both conditions, so neither the lexical identity of the word nor its frequency can be responsible. The subject noun was then reached later in the future, 1432 ms (SE 11.5) against 1419 ms (SE 11.5) (β = 0.037, 95% CrI [0.011, 0.063], P(dir) = > 0.99), was itself produced faster, 122 ms/phone (SE 1.2) against 127 ms/phone (SE 1.2) (β = −0.049, 95% CrI [−0.074, −0.023], P(dir) = > 0.99), and the verb was drawn out, 101 ms/phone (SE 0.9) against 101 ms/phone (SE 1.0) (β = 0.080, 95% CrI [0.048, 0.114], P(dir) = > 0.99). The raw means for verb duration are almost identical across the two tenses, and the model separates them only once frequency is controlled: future trials carry the bare form, which is far more frequent than the past form and therefore shorter, and that difference masks the effect of tense in the unadjusted means.

Repeating the tense from the previous trial did not speed anything up. If anything the numerically primed trials were slower rather than faster, and no main effect of tense repetition was reliable on any measure, including utterance onset (β = 0.015, 95% CrI [−0.015, 0.045], P(dir) = 0.84) and the onset of the subject noun (β = 0.012, 95% CrI [−0.014, 0.038], P(dir) = 0.82).

Repetition did, however, interact with tense, and it did so early. At the onset of the subject noun, repeating the tense delayed the future (1446 ms (SE 16.8) primed against 1417 ms (SE 15.6) unprimed) while the past ran in the opposite direction (1405 ms (SE 15.4) primed against 1433 ms (SE 17.2) unprimed). The interaction is in the predicted direction and most of the posterior supports it, though the interval still includes zero (β = 0.023, 95% CrI [−0.018, 0.064], P(dir) = 0.87). The same asymmetry appears in whether speakers paused before the tense-bearing word at all: tense repetition made a pause less likely in the future than in the past (β = 0.263, 95% CrI [−0.083, 0.608], P(dir) = 0.93), which is the strongest interaction anywhere in the experiment.

Neither interaction survives past the subject noun. The duration of the noun (β = 0.018, 95% CrI [−0.023, 0.059], P(dir) = 0.81) and the duration of the verb (β = 0.001, 95% CrI [−0.040, 0.042], P(dir) = 0.52) separate the conditions by about a millisecond, with the posterior split almost evenly around zero on the verb. Repeating a tense therefore leaves its trace before the verb is reached and nothing at the verb itself, which is where an effect of selecting the tense feature, rather than of retrieving the form that spells it out, should appear. The effect is a lean rather than an established result: with 46 participants the sample is no longer the limiting factor, so the width of the interval reflects the size of the effect.

4 Experiment 2: Morphophonology

The companion to Experiment 1. Here the tense is held constant and the form is varied instead, so that what repeats across trials is not the choice of tense but the way the past is spelled out on the verb. If selecting a tense feature and retrieving its form are separate operations, repetition of the form should act at a different point in the utterance from repetition of the tense.

4.1 Prediction

Repetition should act near the verb. The form of the past tense is not needed until the verb is articulated, so preparing it should leave the time to begin speaking unchanged and appear on the words immediately around the verb. The effect should be larger for irregular verbs, whose past forms have to be retrieved individually, than for regular verbs, which share one ending. The clearest place to see this is the interaction between regularity and repetition, on the duration of the subject noun and on whether the speaker pauses before reaching the verb.

4.2 Participants

41 participants are analysed here, recruited through the same University of Maryland SONA subject pool as Experiment 1 and credited for participation. No participant took part in more than one experiment. Approval, consent and anonymisation were as described for Experiment 1.

4.3 Materials

The characters and the eighteen verbs are those of Experiment 1, in the same three blocks. Ten of the twenty verbs form their past irregularly (*blow, build, dig, drink, eat, read, ride, shake, spin, sweep*) and the rest are regular. Each block holds exactly three of each, so regularity can repeat or switch from trial to trial without being confounded with block.

Every response is in the past tense:

(3) The Pirate spun a top.

4.4 Procedure

The session ran as in Experiment 1, with the same nine blocks in three meta-blocks and the same split between a familiarisation phase and the twelve production trials that follow it. The familiarisation is shorter here: the six verbs of the block are presented one at a time with audio, but nothing places them in time, because every event is past. Blocks are separated by breaks, and the session again yields 108 recorded sentences per participant. Figure 5 gives the block.

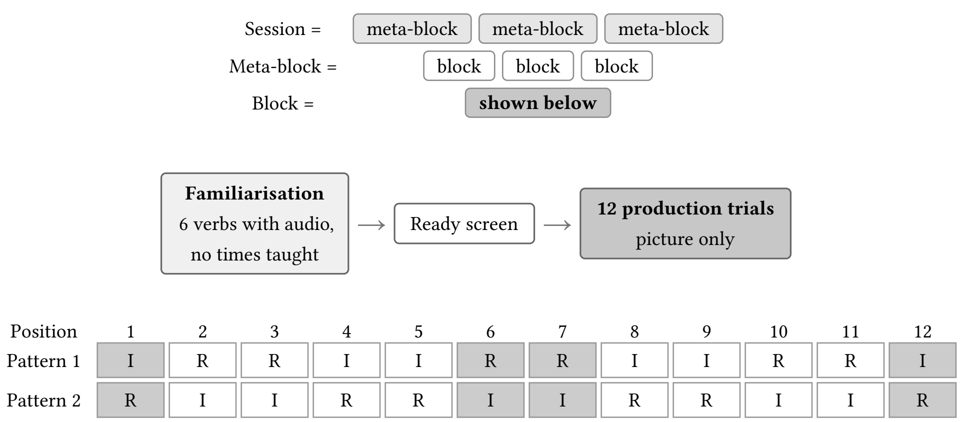


Figure 5: One block of Experiment 2, which is one ninth of the session: three blocks make a meta-block and three meta-blocks make the session. I is an irregular verb and R a regular one. Two passes over the same six verbs, one under each order, with the order of the passes randomised; shaded positions are boundary trials. Tense is never taught, since every event is past. The session, the trial timings and the randomisation are otherwise those of Experiment 1.

Which pattern runs first is randomised per block, block order is shuffled within each meta-block, the assignment of characters to verbs rotates across meta-blocks, and the ordering avoids placing the same character on consecutive trials.

A trial is **primed** when its verb forms the past in the same way as the verb on the trial before it and **unprimed** when it does not. As in Experiment 1, the first and last position of each six-trial pattern have no comparable predecessor and are excluded as boundary trials.

4.5 Analysis

The five measures are those of Experiment 1. Each is modelled separately, and the terms of interest are regularity, regularity repetition, and their interaction. Tense plays no role here, since every trial is past. Frequency, the subject noun and articulation rate are controlled. The appendix gives the contrast coding, the priors, the random-effect structure and the convergence checks.

Of 4344 trials, 545 (13%) failed the utterance screen and 1448 were boundary trials, leaving **2535 trials** from 41 participants.

The model estimates each measure separately. The measures are the time from the start of the trial to the first word, the time to the subject noun, the duration of the subject noun per phone, the duration of the verb per phone, and the interval between the subject noun and the verb.

How to read the signs. A positive coefficient means longer or later onsets and durations; a negative one means shorter or earlier. Predictors are sum coded at ±0.5, so each estimate is the difference between its two levels: Regularity +0.5 = irregular, −0.5 = regular; Regularity priming +0.5 = form-primed, −0.5 = form-unprimed. An interaction is the difference between two such differences.

Speakers reach the verb without pausing on most trials, so the interval is modelled in two parts. **Gap length** is how long the pause lasts on the trials where there is one. **No pause lik.** is how likely the speaker is to run straight into the verb; a positive coefficient there means fewer pauses. The first part is on the log scale in milliseconds, the second on the log-odds scale.

4.6 Results

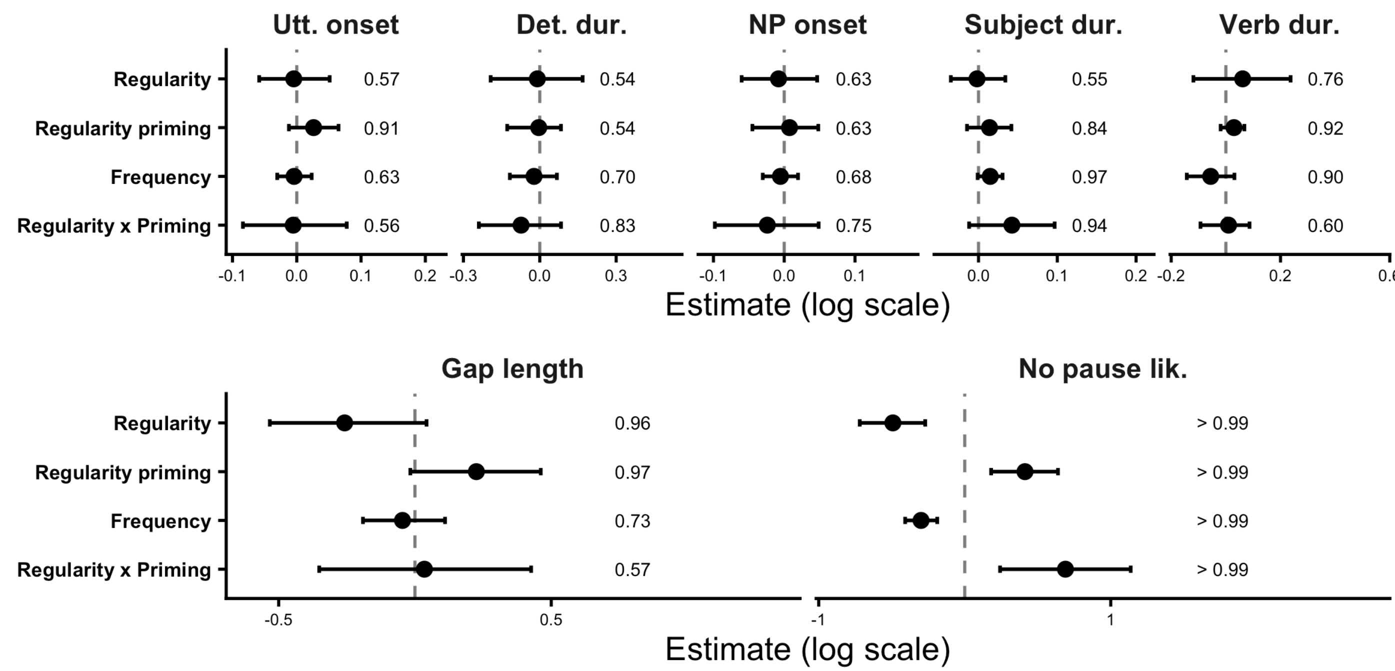


Figure 6: Experiment 2: population-level coefficients of the reported model. Points are posterior medians, bars 95% credible intervals; the dashed line marks zero. The figure at the right of each panel is the proportion of the posterior on one side of zero.

Regularity on its own did almost nothing (Figure 2, middle row). Irregular and regular verbs did not differ at utterance onset (β = −0.005, 95% CrI [−0.058, 0.051], P(dir) = 0.57), at the onset of the subject noun (β = −0.008, 95% CrI [−0.060, 0.047], P(dir) = 0.63), or in the duration of the subject noun (β = −0.002, 95% CrI [−0.035, 0.034], P(dir) = 0.55). The verb itself was numerically longer when irregular, but the interval is wide and includes zero (β = 0.061, 95% CrI [−0.119, 0.237], P(dir) = 0.76).

Repeating the way the past is formed did leave a trace, and it appeared at the verb. Verbs were drawn out when the preceding trial formed its past the same way, 89 ms/phone (SE 1.0) against 97 ms/phone (SE 1.1), with most of the posterior on that side (β = 0.030, 95% CrI [−0.019, 0.068], P(dir) = 0.92). This is where the account expects form repetition to act: the shape of the past is not needed until the verb is spoken.

The same repetition also delayed the start of the utterance, 1082 ms (SE 10.3) against 1083 ms (SE 10.6) (β = 0.026, 95% CrI [−0.012, 0.065], P(dir) = 0.91), which the account does not predict. Nothing about the form of a verb that has not been reached should govern when a speaker begins to talk, so this effect sits awkwardly with the claim that form repetition acts late. Both effects are leans rather than established results, and they are the same size, so the experiment does not on its own separate an early locus from a late one.

The interaction between regularity and repetition was expected to be clearest on the words leading up to the verb, and the largest one in the experiment is indeed on the duration of the subject noun (β = 0.042, 95% CrI [−0.012, 0.097], P(dir) = 0.94). Repetition lengthened the noun more before an irregular verb than before a regular one, which is the direction predicted if an irregular past has to be retrieved individually while regulars share one ending. It does not reach conventional credibility, and no comparable interaction appears at utterance onset (β = −0.006, 95% CrI [−0.084, 0.078], P(dir) = 0.56) or on the verb itself (β = 0.010, 95% CrI [−0.093, 0.087], P(dir) = 0.60).

5 Experiment 3: Conceptual task

The control for the other two. Experiments 1 and 2 both find their effects by repeating something across trials, and both require speech. This experiment repeats the same decision about time without asking for any utterance, so it tests whether a repetition effect needs linguistic material to act on at all.

5.1 Prediction

Repetition should have no effect here. If the repetition effects in the production experiments come from reusing linguistic material, then a task that requires no verb form to be produced gives nothing to reuse. Participants decide which time a picture refers to and press a key. Deciding about the future may take longer than deciding about the past, but repeating a decision about the same time should not speed up the next one, and the effect of repetition should not differ between the two tenses. A repetition effect here would mean the effects in the other two experiments need not be linguistic at all.

5.2 Participants

106 participants are analysed here, recruited through the same University of Maryland SONA subject pool as the production experiments. Approval, consent and anonymisation were as described for Experiment 1; nothing was recorded here, so the session produced no audio.

5.3 Materials

The same three characters and the same twenty verbs as Experiments 1 and 2, with each character paired with each verb and a time assigned to that pairing. Because nothing is spoken, the verb enters only through the picture and the learned pairing, not through a produced form.

5.4 Procedure

Participants learned the character, verb and time pairings, then made speeded keypress decisions about which time a picture referred to, pressing **C** for past and **M** for future. Trials were arranged in twelve-trial blocks on the same scheme as the production experiments, so that tense repeats or switches from one trial to the next in the same way. A trial is **primed** when its tense matches the previous trial's and **unprimed** when it switches, and positions 1, 6, 7 and 12 of each block are boundary trials, as in Figure 3. Figure 7 gives the trial.

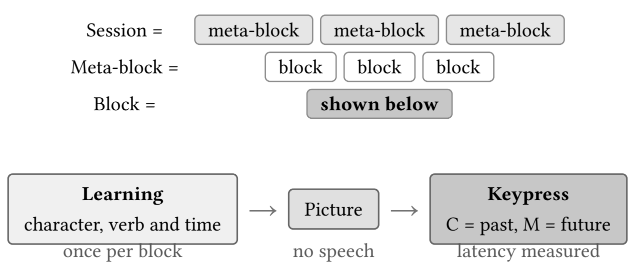


Figure 7: One trial of Experiment 3, and the learning that precedes it. A block teaches its pairings once and then runs twelve trials; three blocks make a meta-block and three meta-blocks make the session, as in the production experiments. The pairings are learned first, as in the production experiments, and the trial itself asks only for a keypress. Nothing is spoken, so the articulatory measures of Experiments 1 and 2 do not apply.

5.5 Analysis

The single measure is the latency of the keypress reporting the decision. The terms of interest are tense, tense repetition, and their interaction. No speech is recorded, so the utterance screen and the articulatory controls used in the other two experiments do not apply.

Of 11880 trials, 3960 boundary trials (positions 1, 6, 7 and 12 of each 12-trial block) were excluded. A further 493 trials fell outside the 200–5000 ms response window and were dropped as anticipations or disengagement: the raw responses run from 0 ms to 173 s, and the upper tail distorts both the means and the lognormal fit. That leaves **7427 trials**.

Experiment 3 has a single dependent measure, the latency of the keypress reporting the decision, so one model is fitted rather than a set over measures. The task involves no speech, so none of the articulatory measures used in Experiments 1 and 2 apply here.

How to read the signs. A positive coefficient means longer or later decision latencies; a negative one means shorter or earlier. Predictors are sum coded at ±0.5, so each estimate is the difference between its two levels: Tense +0.5 = future, −0.5 = past; Tense priming +0.5 = tense-primed, −0.5 = tense-unprimed; Regularity priming +0.5 = form-primed, −0.5 = form-unprimed. An interaction is the difference between two such differences.

5.6 Results

Regularity repetition is also tracked. The two are not independent: regularity repeats on 50% of trials where tense repeats but 42% of trials where it switches, so the two priming effects are estimated jointly in one model rather than separately; the cell means are in the appendix.

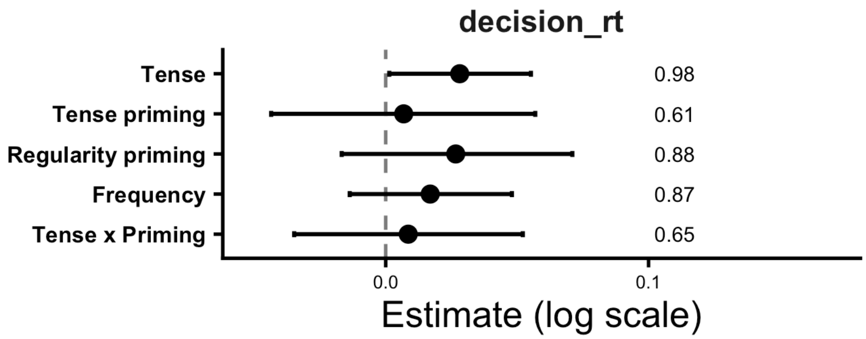


Figure 8: Experiment 3: population-level coefficients of the reported model for decision latency. Points are posterior medians, bars 95% credible intervals. The figure at the right is the proportion of the posterior on one side of zero.

Decisions were slower for future than for past (Figure 2, bottom row). Future trials averaged 1125 ms (SE 11.4) and past trials 1099 ms (SE 11.4). The model supports this difference ($\beta = 0.028$, 95% CrI [0.001, 0.055], P(dir) = 0.98).

Tense repetition did not affect decision times. Primed trials averaged 1108 ms (SE 11.2) and unprimed trials 1116 ms (SE 11.6), and the model gives $\beta = 0.007$, 95% CrI [−0.044, 0.057], P(dir) = 0.61. The interaction between tense and tense repetition is absent ($\beta = 0.009$, 95% CrI [−0.035, 0.052], P(dir) = 0.65), with the posterior centred on zero.

This is what we expected. The conceptual task requires participants to retrieve which time a picture refers to and press a key, with no verb form to produce. Deciding about the future takes longer than deciding about the past, but repeating a decision about the same time does not make the next one faster.

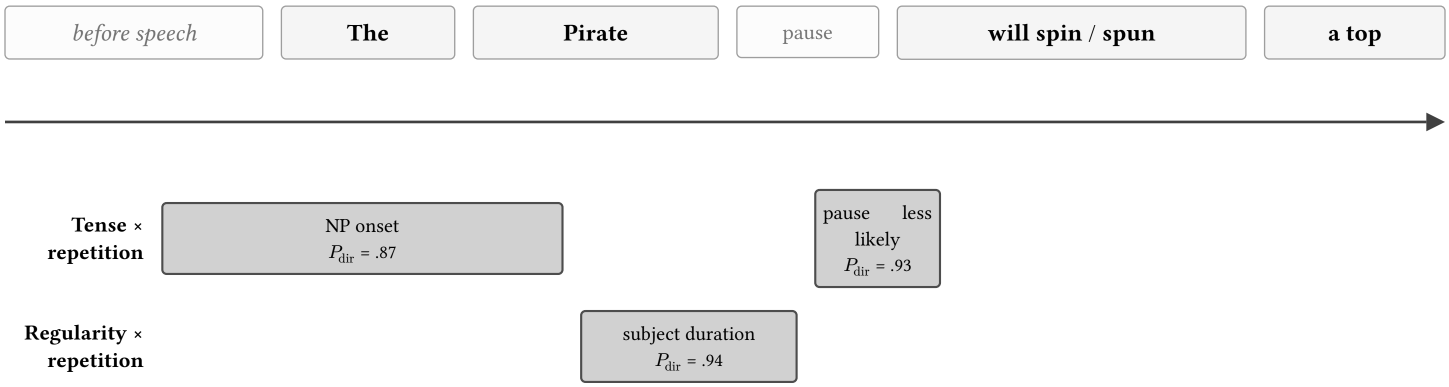
6 What the three experiments show

Repeating a tense left its mark before the verb was reached and nothing at the verb itself. The interaction between tense and tense repetition ran in the predicted direction at the onset of the subject noun, and repetition made a pause before the tense-bearing word less likely in the future than in the past; neither the duration of the subject noun nor the duration of the verb separated the conditions. Tense itself was already visible on the determiner, a word identical in both conditions, which is the earliest point at which it could show up at all. That is the pattern H1 predicts and H2 does not.

Repeating the form gave a weaker and less well-placed result. The interaction between regularity and repetition, which is the term that separates a form that has to be retrieved individually from one that shares an ending, fell on the duration of the subject noun and not on the verb. Nothing comparable appeared at the verb itself. Experiment 3 produced no interaction anywhere, so whatever the production experiments are picking up is not a generic effect of repeating a decision.

Main effects of repetition are left out of this summary. Both experiments show some, but a main effect says only that repetition changed a measure, not when the information was planned, and the two accounts make the same prediction about it. Only the interactions distinguish them.

Figure 9 places these results on the same timeline as the predictions.



Only interactions are shown: a main effect of repetition, on its own, does not tell the two accounts apart. Tense repetition acts to the left of the verb and not on it, as H1 predicts.

Figure 9: Where the effects actually fell, on the timeline of `@fig-predictions`{=typst}. Only the interactions are shown, since these are what the two accounts disagree about; each band gives the proportion of the posterior on one side of zero. Every one is a lean rather than a credible effect, as no interval excluded zero. Regions left blank are ones where the measure was taken and no interaction was found. Experiment 3 is not shown: it produced no interaction anywhere.

A1. Model coefficients

The tables behind the coefficient figures in each Results section. Estimates are on the log scale, so a coefficient is a proportional change; positive means slower or longer. Bold marks terms with P(dir) at or above the threshold set in the header of this document. Terms held in the models as controls but left out of the figures are also left out here.

Experiment 1: Morphosyntax

Measure	Term	Estimate	95% CrI	P(dir)
Utt. onset	Tense priming	0.015	[−0.015, 0.045]	0.84
	Tense	0.010	[−0.020, 0.039]	0.75
	Tense x Priming	0.014	[−0.038, 0.065]	0.71
	Frequency	0.004	[−0.011, 0.020]	0.69
	Articulation rate	**0.024**	**[0.010, 0.038]**	**> 0.99**
Det. dur.	Tense priming	−0.007	[−0.070, 0.055]	0.60
	Tense	**0.139**	**[0.078, 0.201]**	**> 0.99**
	Tense x Priming	−0.012	[−0.097, 0.072]	0.61
	Frequency	0.003	[−0.050, 0.059]	0.54
	Articulation rate	**-0.446**	**[−0.473, −0.418]**	**> 0.99**
NP onset	Tense priming	0.012	[−0.014, 0.038]	0.82
	Tense	**0.037**	**[0.011, 0.063]**	**> 0.99**
	Tense x Priming	**0.023**	**[−0.018, 0.064]**	**0.87**
	Frequency	−0.001	[−0.020, 0.018]	0.55
	Articulation rate	**-0.080**	**[−0.093, −0.068]**	**> 0.99**
Subject dur.	Tense priming	−0.008	[−0.033, 0.017]	0.74
	Tense	**-0.049**	**[−0.074, −0.023]**	**> 0.99**
	Tense x Priming	0.018	[−0.023, 0.059]	0.81
	Frequency	0.006	[−0.007, 0.020]	0.81

Measure	Term	Estimate	95% CrI	P(dir)
	Articulation rate	**0.032**	**[0.020, 0.045]**	**> 0.99**
Verb dur.	Tense priming	0.011	[−0.019, 0.040]	0.77
	Tense	**0.080**	**[0.048, 0.114]**	**> 0.99**
	Tense x Priming	0.001	[−0.040, 0.042]	0.52
	Frequency	**-0.095**	**[−0.125, −0.067]**	**> 0.99**
	Articulation rate	0.006	[−0.007, 0.018]	0.81
Gap length	Tense priming	−0.018	[−0.175, 0.141]	0.59
	Tense	−0.048	[−0.207, 0.106]	0.73
	Tense x Priming	0.002	[−0.234, 0.243]	0.51
	Frequency	−0.009	[−0.100, 0.095]	0.57
	Articulation rate	**0.090**	**[0.013, 0.167]**	**0.99**
No pause lik.	Tense priming	−0.070	[−0.247, 0.105]	0.78
	Tense	**0.281**	**[0.083, 0.478]**	**> 0.99**
	Tense x Priming	**0.263**	**[−0.083, 0.608]**	**0.93**
	Frequency	**-0.170**	**[−0.267, −0.076]**	**> 0.99**
	Articulation rate	**0.121**	**[0.017, 0.229]**	**0.99**

Table 1: Experiment 1, reported model (spec 5). Positive = slower/longer. Bold marks terms with P(dir) at or above 0.85.

Experiment 2: Morphophonology

Measure	Term	Estimate	95% CrI	P(dir)
Utt. onset	**Regularity priming**	**0.026**	**[−0.012, 0.065]**	**0.91**
	Regularity	−0.005	[−0.058, 0.051]	0.57
	Regularity x Priming	−0.006	[−0.084, 0.078]	0.56
	Frequency	−0.004	[−0.030, 0.023]	0.63
	Articulation rate	**0.094**	**[0.066, 0.121]**	**> 0.99**
Det. dur.	Regularity priming	−0.004	[−0.128, 0.083]	0.54
	Regularity	−0.010	[−0.193, 0.168]	0.54
	Regularity x Priming	−0.074	[−0.239, 0.083]	0.83
	Frequency	−0.023	[−0.118, 0.067]	0.70
	Articulation rate	**-0.763**	**[−0.813, −0.710]**	**> 0.99**
NP onset	Regularity priming	0.007	[−0.045, 0.048]	0.63
	Regularity	−0.008	[−0.060, 0.047]	0.63
	Regularity x Priming	−0.024	[−0.098, 0.049]	0.75
	Frequency	−0.005	[−0.030, 0.020]	0.68
	Articulation rate	**-0.121**	**[−0.143, −0.099]**	**> 0.99**
Subject dur.	Regularity priming	0.014	[−0.015, 0.042]	0.84
	Regularity	−0.002	[−0.035, 0.034]	0.55

Measure	Term	Estimate	95% CrI	P(dir)
	Regularity x Priming	**0.042**	**[−0.012, 0.097]**	**0.94**
	Frequency	**0.015**	**[−0.001, 0.031]**	**0.97**
	Articulation rate	**0.012**	**[−0.009, 0.035]**	**0.87**
Verb dur.	**Regularity priming**	**0.030**	**[−0.019, 0.068]**	**0.92**
	Regularity	0.061	[−0.119, 0.237]	0.76
	Regularity x Priming	0.010	[−0.093, 0.087]	0.60
	Frequency	**-0.056**	**[−0.143, 0.031]**	**0.90**
	Articulation rate	−0.009	[−0.033, 0.014]	0.78
Gap length	**Regularity priming**	**0.225**	**[−0.017, 0.462]**	**0.97**
	Regularity	**-0.258**	**[−0.533, 0.043]**	**0.96**
	Regularity x Priming	0.035	[−0.351, 0.426]	0.57
	Frequency	−0.046	[−0.191, 0.110]	0.73
	Articulation rate	**0.111**	**[−0.071, 0.288]**	**0.89**
No pause lik.	**Regularity priming**	**0.413**	**[0.181, 0.639]**	**> 0.99**
	Regularity	**-0.492**	**[−0.719, −0.271]**	**> 0.99**
	Regularity x Priming	**0.690**	**[0.242, 1.137]**	**> 0.99**
	Frequency	**-0.299**	**[−0.408, −0.188]**	**> 0.99**
	Articulation rate	**0.323**	**[0.098, 0.548]**	**> 0.99**

Table 2: Experiment 2, reported model (spec 5). Positive = slower/longer. Bold marks terms with P(dir) at or above 0.85.

Experiment 3: Conceptual task

Measure	Term	Estimate	95% CrI	P(dir)
decision_rt	Tense priming	0.007	[−0.044, 0.057]	0.61
	Regularity priming	**0.027**	**[−0.017, 0.071]**	**0.88**
	Tense	**0.028**	**[0.001, 0.055]**	**0.98**
	Tense x Priming	0.009	[−0.035, 0.052]	0.65
	Frequency	**0.017**	**[−0.014, 0.048]**	**0.87**

Table 3: Experiment 3, reported model (spec 5). Positive = slower/longer. Bold marks terms with P(dir) at or above 0.85.

A2. Condition means

The tables behind the figures in the Descriptive results section.

Experiment 1: Morphosyntax

Measure	Condition	Prime status	N	Mean	SE	Measure	Condition	Prime status	N	Mean	SE
Utt. onset	future	primed	667	1243	13.2	Subj. dur.	future	primed	672	122	1.7
	future	unprimed	675	1207	12.7		future	unprimed	680	123	2.0
	past	primed	654	1204	13.7		past	primed	656	126	1.7
	past	unprimed	651	1211	13.5		past	unprimed	653	128	1.9
Det. dur.	future	primed	672	182	8.3	Verb dur.	future	primed	672	104	1.3

Measure	Condition	Prime status	N	Mean	SE	Measure	Condition	Prime status	N	Mean	SE
	future	unprimed	680	180	7.9		future	unprimed	680	98	1.3
	past	primed	656	169	6.4		past	primed	632	106	1.5
	past	unprimed	653	181	9.6		past	unprimed	629	97	1.4
NP onset	future	primed	672	1446	16.5	Gap	future	primed	672	72	6.4
	future	unprimed	680	1417	15.7		future	unprimed	680	89	8.4
	past	primed	656	1405	15.3		past	primed	656	94	8.5
	past	unprimed	653	1433	16.8		past	unprimed	653	96	9.0

Table 4: Experiment 1: timing measures by tense and tense repetition. Onsets and the gap are in milliseconds; subject-noun and verb durations are milliseconds per phoneme. SE is the Morey (2008) within-subject standard error.

Experiment 2: Morphophonology

Measure	Condition	Prime status	N	Mean	SE	Measure	Condition	Prime status	N	Mean	SE
Utt. onset	IRREGULAR	primed	612	1077	14.9	Subj. dur.	IRREGULAR	primed	630	115	1.8
	IRREGULAR	unprimed	635	1077	14.3		IRREGULAR	unprimed	649	116	2.2
	REGULAR	primed	622	1087	13.3		REGULAR	primed	642	114	1.7
	REGULAR	unprimed	598	1089	14.7		REGULAR	unprimed	614	120	2.3
Det. dur.	IRREGULAR	primed	630	256	13.4	Verb dur.	IRREGULAR	primed	550	87	1.3
	IRREGULAR	unprimed	649	246	14.2		IRREGULAR	unprimed	649	100	1.5
	REGULAR	primed	642	258	12.9		REGULAR	primed	642	90	1.3
	REGULAR	unprimed	614	234	11.0		REGULAR	unprimed	602	94	1.4
NP onset	IRREGULAR	primed	630	1329	18.5	Gap	IRREGULAR	primed	630	36	5.4
	IRREGULAR	unprimed	649	1323	19.4		IRREGULAR	unprimed	649	40	5.3
	REGULAR	primed	642	1345	18.0		REGULAR	primed	642	46	6.7
	REGULAR	unprimed	614	1311	18.2		REGULAR	unprimed	614	39	6.2

Table 5: Experiment 2: timing measures by regularity and regularity repetition. Onsets and the gap are in milliseconds; subject-noun and verb durations are milliseconds per phoneme. SE is the Morey (2008) within-subject standard error.

Experiment 3: Conceptual task

Tense	Prime status	N	Mean RT (ms)	SE
future	primed	1855	1126	16.1
future	unprimed	1860	1123	16.1
past	primed	1859	1090	15.6
past	unprimed	1853	1109	16.7

Table 6: Experiment 3: decision RT by tense and tense repetition. SE is the standard error of the mean.

Regularity prime	Tense prime	N	Mean RT (ms)	SE
primed	primed	1860	1091	15.1
primed	unprimed	1539	1183	19.2
unprimed	primed	1854	1125	16.6
unprimed	unprimed	2174	1069	14.3

Table 7: Experiment 3: decision RT by regularity repetition and tense repetition, showing how far the two priming variables co-vary.

A3. By-item effects

By-verb deviations from the intercept in the reported model of each experiment. A verb whose interval excludes zero behaves differently from the rest of the item set.

Experiment 1: Morphosyntax

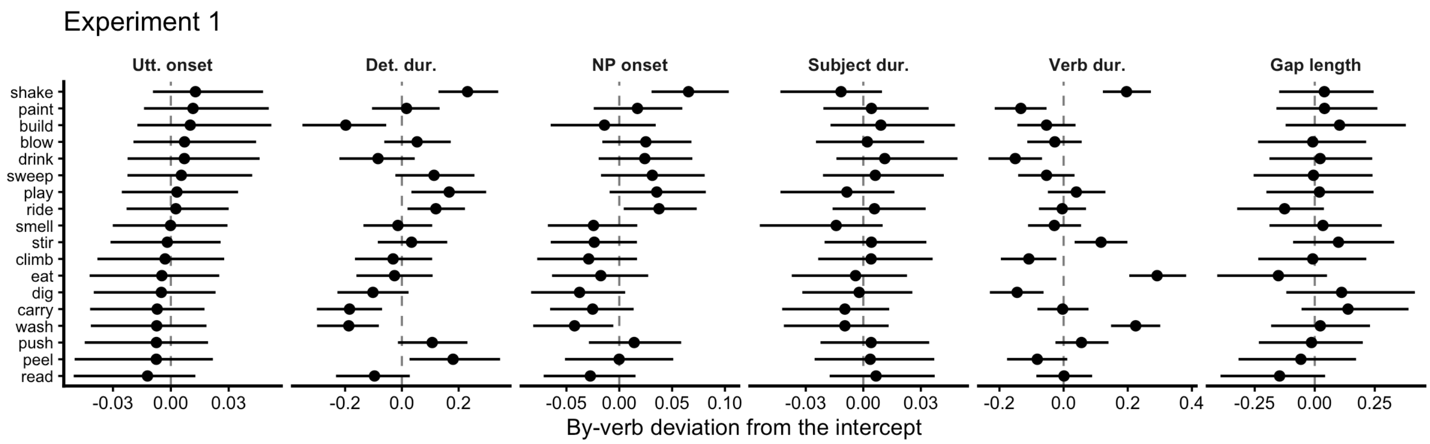


Figure 10: Experiment 1: by-verb deviations from the population intercept in the reported model. A verb whose interval excludes zero behaves differently from the rest of the item set.

Experiment 2: Morphophonology

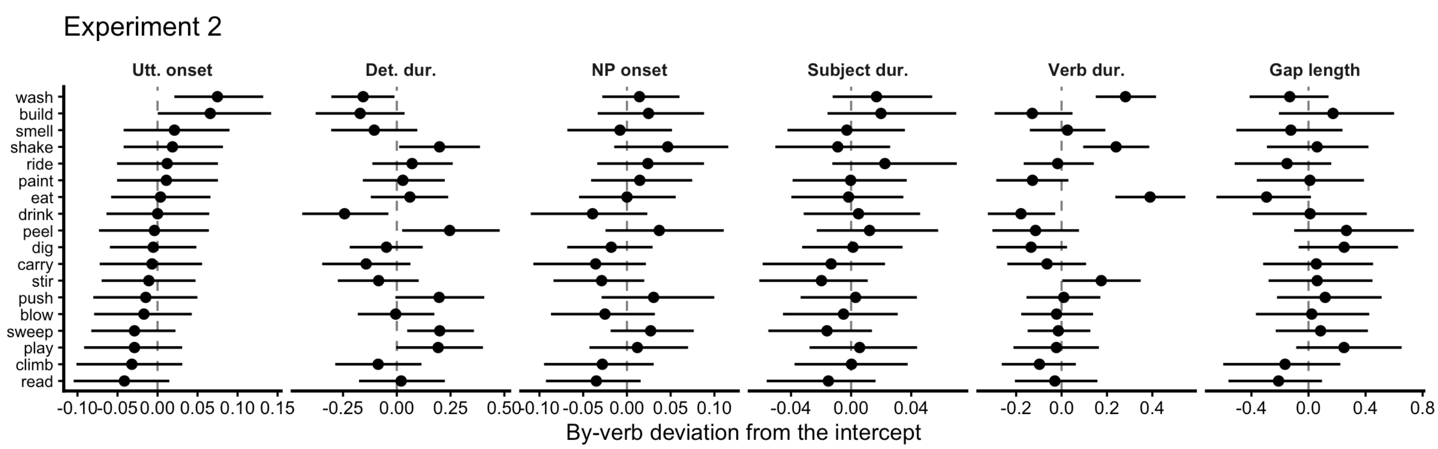


Figure 11: Experiment 2: by-verb deviations from the population intercept in the reported model.

Experiment 3: Conceptual task

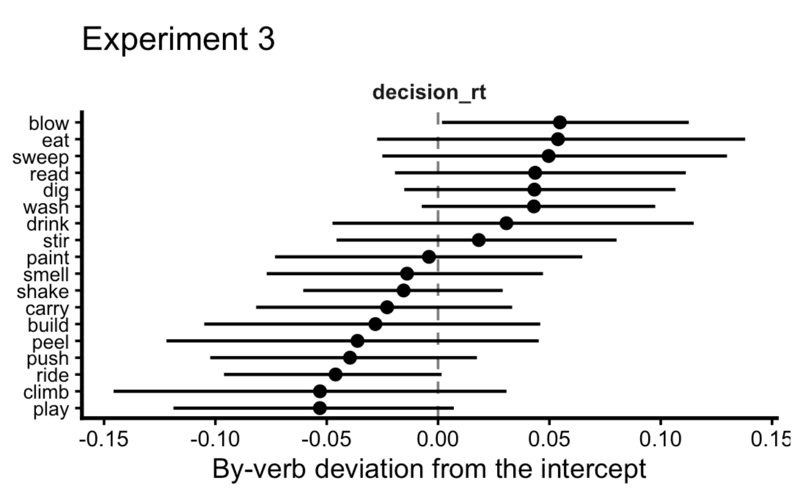


Figure 12: Experiment 3: by-verb deviations from the population intercept in the reported model.

A4. Model comparison and checks

Model comparison

Models were compared by leave-one-out cross-validation. Differences are relative to the best model for that measure; a difference smaller than about twice its standard error is not meaningful.

Model comparison is switched off. Set `run_loo <- TRUE` to compute it.

Convergence

No divergent transitions, R-hat at or below 1.01, and effective sample sizes in the thousands for every model reported.

measure	divergences	max_rhat	min_ess_bulk	min_ess_tail	ok
decision_rt	0	1.003	1116	2435	TRUE

measure	divergences	max_rhat	min_ess_bulk	min_ess_tail	ok
Utt. onset	0	1.005	946	2114	TRUE
Det. dur.	0	1.002	1973	2294	TRUE
NP onset	0	1.002	1814	2215	TRUE
Subject dur.	0	1.005	1068	1839	TRUE
Verb dur.	2	1.005	1055	1693	FALSE
Gap length	0	1.002	1839	1699	TRUE

measure	divergences	max_rhat	min_ess_bulk	min_ess_tail	ok
Utt. onset	0	1.003	1129	2359	TRUE
Det. dur.	0	1.004	1187	2248	TRUE
NP onset	0	1.003	1585	2393	TRUE
Subject dur.	0	1.006	1012	1826	TRUE
Verb dur.	0	1.005	1164	1151	TRUE
Gap length	0	1.002	2163	3164	TRUE

Posterior predictive checks

Each panel compares the observed distribution of a measure (dark line) against data simulated from the fitted model (light lines).

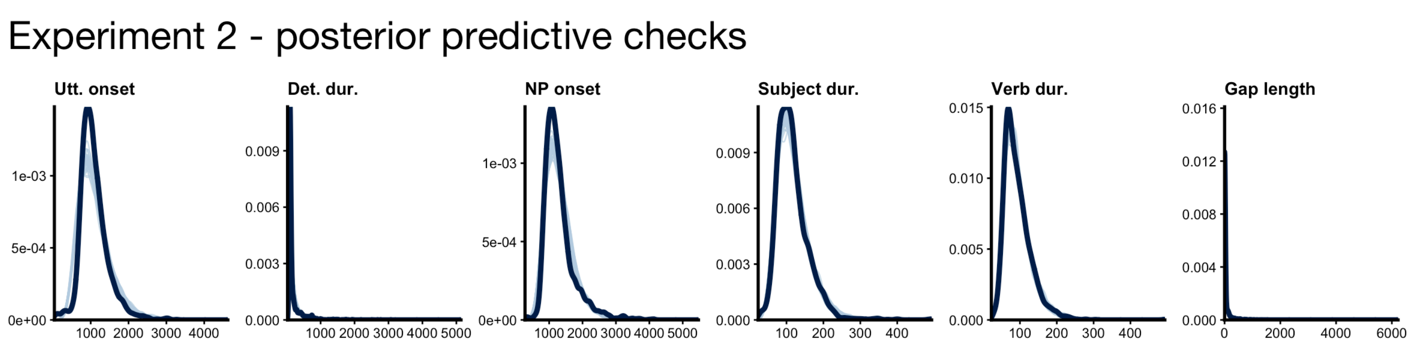


Figure 13: Experiment 2: posterior predictive checks. Simulated datasets from the fitted model against the observed distribution of each measure.

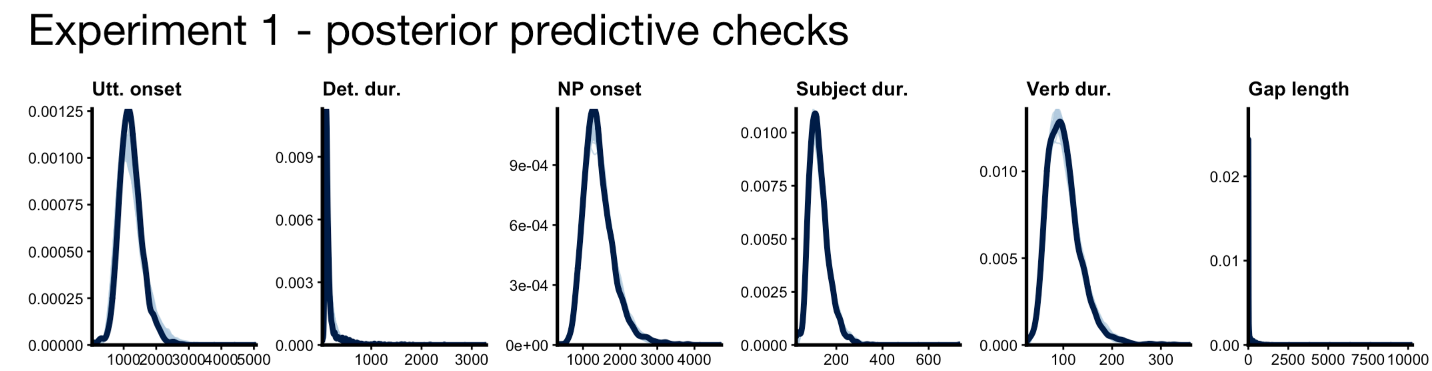


Figure 14: Experiment 1: posterior predictive checks. Simulated datasets from the fitted model against the observed distribution of each measure.

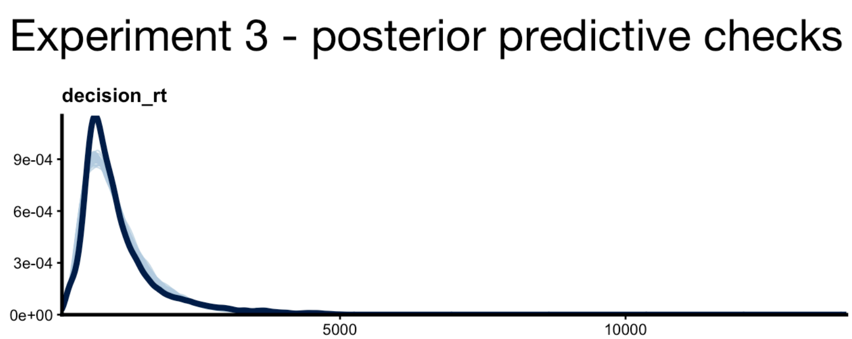


Figure 15: Experiment 3: posterior predictive checks for decision latency.

A5. Statistical models

All models were fitted with brms (Stan), four chains of 4000 iterations with the first 1000 discarded, `adapt_delta` = 0.95.

Contrast coding

All categorical predictors are sum coded at ± 0.5, so each coefficient is the difference between the two levels and the intercept is the grand mean.

predictor	+0.5	−0.5	used in
Tense	future	past	Experiments 1 and 3
Regularity	irregular	regular	Experiment 2
Tense priming	primed	unprimed	Experiments 1 and 3
Regularity priming	primed	unprimed	all three

In Experiment 2 regularity priming is the manipulation and appears as *Priming* in the tables. In Experiments 1 and 3 the manipulation is tense priming, and regularity priming enters only as a control.

The subject noun (Chef, Pirate, Wizard) is treated as a factor with Chef as the reference level. Frequency is the rank of the produced word form in the SUBTLEX Zipf scale, from least to most frequent, centred and divided by its standard deviation. Ranking rather than the Zipf value itself keeps a small number of very frequent verbs from dominating an otherwise flat predictor. Articulation rate is the number of phones per second on that trial, computed over the words that are not being measured, and is also centred and scaled.

Response distributions

Onsets and durations are positive and right skewed and were fitted with a shifted lognormal distribution. The shift estimates a floor below which no response occurs.

The interval between the subject noun and the verb is zero on 80% of trials in Experiment 2 and 66% in Experiment 3, because speakers often move into the verb without pausing. A single continuous distribution describes this badly, so it was fitted with a hurdle lognormal: one part gives the probability of producing no pause, the other the duration of the pause when there is one. Both parts take the same predictors, which allows repetition to change how often speakers pause separately from how long they pause. Coefficients labelled as being on the probability of no pause come from the first part, where a positive value means fewer pauses.

Model structure
Every measure was fitted with the same structure. For Experiments 1 and 3,

$$\text{measure} \sim \text{Tense} \times \text{Tense priming} + \text{Regularity priming} + \text{Frequency} + \text{Character} + \text{Rate}$$

$$+(1 + \text{Tense} \times \text{Tense priming} + \text{Regularity priming} \mid \text{participant}) + (1 + \text{Tense priming} \mid \text{verb})$$

and for Experiment 2,

$$\text{measure} \sim \text{Regularity} \times \text{Priming} + \text{Frequency} + \text{Character} + \text{Rate}$$

$$+(1 + \text{Regularity} \times \text{Priming} \mid \text{participant}) + (1 + \text{Priming} \mid \text{verb})$$

Experiment 1 has no recordings and therefore no rate term.

Each measure was fitted with the same population-level terms: the design factor (tense or regularity), repetition, their interaction, frequency, the subject noun, articulation rate, and, in Experiments 1 and 3, form repetition as a control. Random effects are by participant, with slopes for the design factor, repetition and their interaction, and by verb, with a slope for repetition. Frequency, the subject noun and articulation rate vary between items or within a trial and take no by-participant slope.

Regularity priming is included because it is not independent of tense priming in these materials. The form repeats on 50% of trials where tense repeats and 41% of trials where it switches, so leaving it out would let part of a form-repetition effect appear as a tense priming effect. It is never entered into an interaction.

Priors

parameter	prior
intercept	Normal(0, 1)
population-level coefficients	Normal(0, 0.3)
random-effect standard deviations	Normal(0, 0.3)
random-effect correlations	LKJ(2)
hurdle intercept and coefficients	Normal(0, 1)

Coefficients are on the log scale, so Normal(0, 0.3) places most of its mass on effects smaller than a factor of two and rules out nothing the data could show.

Reporting
Each coefficient is reported as its posterior median, a 95% credible interval, and P(dir), the proportion of the posterior lying on the same side of zero as the median. P(dir) runs from 0.5, meaning the posterior is split evenly and the sign is undetermined, to 1, meaning the whole posterior falls on one side. It is not a significance test and has no threshold attached; we describe an effect as supported when most of the posterior is on one side and the interval is narrow enough to be informative. Values are never printed as 0 or 1, since a finite posterior sample cannot establish either.

Convergence and model checking
Every model reported here converged: no divergent transitions, R-hat at or below 1.01, and bulk effective sample sizes above 400. Posterior predictive checks compare simulated data against the observed distribution for each measure. Posterior contraction, the reduction in variance from prior to posterior, is above 0.94 for the frequency, subject noun and articulation rate terms and between 0.56 and 0.99 for the interactions, so the posteriors are determined by the data rather than by the priors.

Models were compared by leave-one-out cross-validation. Adding the subject noun and articulation rate improves prediction on the duration measures by a wide margin, which is why they are in the reported model: the three subject nouns differ by more than 50 ms per phone, far more than any effect of the manipulation.

A6. Recordings and measurement
Spoken responses were recorded in the browser, one file per trial, and uploaded in blocks. Recordings were converted to 16 kHz mono audio and transcribed automatically. Each transcript was then aligned to its recording with the Montreal Forced Aligner using its English acoustic model and pronunciation dictionary, with pronunciations generated for words outside the dictionary. The alignment gives the start and end time of every word and every phone, and all timing measures are read from it.

Speech onset is the start of the first word. The onset of the subject noun and its duration are taken from the second word. The interval before the verb runs from the end of the subject noun to the start of the verb, and is zero when the two are adjacent. The verb is located by matching the transcript against the form the trial called for rather than by position, since the future adds a word before it.

Durations of the subject noun and the verb are divided by the number of phones the aligner placed inside them. Words differ in length, and without this a longer word would count as slower speech. Articulation rate is the number of phones per second over the remaining words of the sentence, which leaves the two measured words out of their own control variable.

Transcripts were scored against the sentence the trial called for. A response was kept if it named the right character and used the right verb in the right tense. Material after the verb was not scored, since every measure lies at or before the verb. Two adjustments were made after inspecting the transcripts. The past tense of *blow* is a homophone of *blue* and was transcribed as the colour on most trials, so both spellings were accepted. *Blowed* was kept as an error, being a real regularisation rather than a transcription problem. Responses that marked time with an adverbial instead of on the verb, such as *the wizard climbs a ladder in the past*, were excluded, since they leave nothing for the verb to carry.

Participants were excluded if fewer than half their responses met these criteria.